# ASSESSMENT AGENT SPECIFICATION

**Member 2: Evaluation, Knowledge Diagnostics & Blueprint Pipeline**

AGENT 2 OF 5

## 1. AGENT OVERVIEW & CORE PURPOSE

The **Assessment Agent** is an autonomous evaluation pipeline. It moves beyond standard MCQ generation by designing validated examination blueprints, conducting deep diagnostic assessments, isolating exact knowledge gaps, and updating the system's global mastery state.

**Primary Objective**

Generate, validate, and evaluate structured quizzes, exams, and flashcards grounded directly in course materials.

**Diagnostic Core**

Categorize student errors (e.g., calculation error vs. structural misconception) to update topic-level mastery metrics.

## 2. UNIVERSAL SYSTEM DATA SCHEMA CONTEXT

The Assessment Agent is responsible for reading from and updating the `knowledge_state` block:

```
{ "user_profile": { "user_id": "U101" }, "knowledge_state": { "CS101": { "Graphs": { "mastery": 0.32, "last_tested": "2026-10-01", "error_patterns": ["Dijkstra edge weights"] } } }, "assessment_logs": [ { "quiz_id": "Q-99", "score": 0.40, "identified_gaps": ["Graph Traversal"] } ] }
```

## 3. INPUT & OUTPUT SPECIFICATION

| Type | Input Sources / Formats | Outputs / Artifacts Produced |
|---|---|---|
| **Data Payload** | • Course Material Vector Context<br>• User Quiz Responses (Text/MCQ)<br>• Target Bloom's Taxonomy Level | • Validated Quizzes & Flashcard Decks<br>• Detailed Error Analysis & Diagnostic Feedback<br>• Updated Topic Mastery Scores |
| **Control Signals** | • Assessment Trigger Signal from **Adaptive Manager** or **Tutor Agent** | • Diagnostic Alert Signal sent to Adaptive Manager when severe mastery drops occur |

## 4. MULTI-STEP ASSESSMENT WORKFLOW

The Assessment Agent executes a deterministic multi-stage evaluation pipeline:

1. **Blueprint Creation:** Defines test distribution based on current `knowledge_state` weak points.
2. **Generation & Validation:** Synthesizes questions via LLM and executes an internal validation pass to verify clarity, correct answers, and document alignment.
3. **Evaluation & Error Classification:** Grades user submission, distinguishing between minor oversight and fundamental conceptual misunderstanding.
4. **Schema Update:** Computes new mastery score and updates global JSON store.

## 5. INTER-AGENT WORKFLOW & INTEGRATION ARCHITECTURE

The Assessment Agent acts as the analytical feedback authority for the entire system:

**Inbound Interactions**

- **AI Tutor Agent:** Requests diagnostic quiz generation following a tutoring topic completion.
- **Adaptive Manager:** Triggers periodic retention testing based on scheduled reviews.

**Outbound Interactions**

- **Adaptive Manager:** Sends evaluation results (mastery deltas and identified knowledge gaps).
- **Planning Agent:** Feeds weak topic lists to enforce targeted review blocks in schedule.

## 6. DETAILED AGENT COMMUNICATION MATRIX

| Target Agent | Data Shared / Received | Trigger Condition / Purpose |
|---|---|---|
| **Adaptive Manager** | Diagnostic Failure Alerts & Updated Mastery Map | Sent immediately after exam grading to trigger adaptive replanning. |
| **AI Tutor Agent** | Specific Misconception Identifiers & Sub-topic Tags | Directs Tutor Agent on exact concept areas requiring remedial explanation. |
| **Personal Planning Agent** | Required Remediation Topics & Urgency Weights | Pushes high-priority study tasks for failed concepts into student schedule. |

## 7. VALUE PROPOSITION & SYSTEM IMPACT

The Assessment Agent provides the objective benchmark data necessary for true adaptation:

- **Data-Driven Adaptation:** Prevents self-reported user confidence from biasing the system by enforcing verified competence metrics.
- **Grounded Diagnostic Feedback:** Pinpoints precise conceptual failures rather than giving generic right/wrong test scores.
- **Loop Closure:** Provides the quantitative `Measure` step required for the global adaptive execution loop.